SVM notes

November 2026

1 What is Support Vector Machine?

1. problem setting: - we want to separate a set of linear separable data points - unknown target function $f : X \to Y$ that correctly separates the data points and satisfies our goal - set of function hypothesis: $H = \{h|h : X \to Y\}$ 2. Input: Training examples of an unknown target function f 3. Output: Hypothesis $h \in H$ that can maximize the distance between the decision boundary while correctly classifying our data points 4. Should be linear ($Y = \omega^T \cdot \vec{x} + \vec{b}$)

2 Define the margin

The Margin of the dataset $mathbfD$ and decision boundary $\omega \cdot x + b$ is given by:

$$\text{margin}(\mathbf{D}, \omega, b) = \begin{cases} \min_{(x,y)\in\mathbf{D}} y(\omega \cdot x + b) & \text{if } \omega \text{ separates } \mathbf{D} \\ -\infty & \text{otherwise} \end{cases} \tag{3.8}$$

which is the nearest distance between the decision boundary and some points in the dataset $\mathbf{D}$, And the margin for a dataset $\mathbf{D}$ is defined by the largest attainable margin for dataset $mathbfD$:

$$\text{margin}(\mathbf{D}) = \sup_{\boldsymbol{w},b} \text{margin}(\mathbf{D}, \boldsymbol{w}, b)$$

3 How to find it?

Obviously, the distance between the nearest positive data and decision boundary (denoted as: d_+) should equal to the distance between the nearest negative data and the decision boundary (denoted as: d_-). That is because if not, assume our best decision boundary has property $d_- > d_+$, let move the decision boundary towards negative direction, we find the margin increase. So, the old decision boundary is not the best one(contradictory), which suggests $d_+ = d_-$ for the best decision boundary.

Then, let's assume we have two hyperplanes: $\omega \cdot x + b = c_1$ and $\omega \cdot x + b = c_2$. $\omega \cdot x + b = c_1$ touches the positive instance and all positive instances are above this plane (called "plus-plane") and $\omega \cdot x + b = c_2$ touches the negative instance and all negative instances are below this plane (called "minus-plane"). From above discussion, we know $d_+ = \frac{|c_1|}{\|\omega\|} = d_- = \frac{|c_2|}{\|\omega\|}$. Solve this equation we get: $|c_1| = |c_2|$, and we know $c_1 \neq c_2$ (otherwise, plus-plane and the minus-plane will be the same plane). we get: $c_1 = -c_2$. Let's denote $c_1 = c$, so we have:

$$\text{plus-plane: } \omega \cdot x + b = c$$
$$\text{minus-plane: } \omega \cdot x + b = -c$$

Let's further devide both side by c ($c \neq 0$), we get:

$$\text{plus-plane: } \frac{\omega}{c} \cdot x + \frac{b}{c} = 1$$
$$\text{minus-plane: } \frac{\omega}{c} \cdot x + \frac{b}{c} = -1$$

And replacing $\frac{\omega}{c}$ by ω and $\frac{b}{c}$ by b, we get:

$$\text{plus-plane: } \omega \cdot x + b = 1$$
$$\text{minus-plane: } \omega \cdot x + b = -1$$

We call equations in above form: **canonical form**, and it has one interesting property: the margin of this hyperplane (in canaical form) is: $\frac{1}{\|\|}$

That is because: distance $= \frac{|\omega \cdot x + b|}{\|\|} = \frac{1}{\|\|}$. So, we can maximize the margin by minimize the $\|\omega\|$

4 Optimalization

Since minimizing $\|\omega\|$ is same as minimizing $\frac{1}{2}\|\omega\|^2$, we can minimize $\frac{1}{2}\|\omega\|^2$ instead.

So, our optimalization problem is defined below:

$$\begin{aligned} &\text{Minimize} && f(\omega, b) = \frac{\|\omega\|^2}{2} \\ &\text{subject to} && y_n(\omega^T \mathbf{x}_n + b) \geq 1, \quad \forall\, n = 1, \ldots, N \end{aligned}$$

To solve this problem, noticing that x, y, ω, b are continues and bounded, we can directly apply 'Lagrange Multuplier' to find where the multipilers give us minimum. (multipliers are: $\alpha_1, \alpha_2, \ldots, \alpha_N$)

$$\frac{\partial(\frac{1}{2}\|\omega\|^2)}{\partial \omega} = \sum_{i=1}^{N} \alpha_i \frac{\partial(1 - y_n(\omega^T \mathbf{x}_n + b))}{\partial \omega}$$
$$\frac{\partial(\frac{1}{2}\|\omega\|^2)}{\partial b} = \sum_{i=1}^{N} \alpha_i \frac{\partial(1 - y_n(\omega^T \mathbf{x}_n + b))}{\partial b}$$

and our problem becomes (if you want to know why this holds, please refer to /Basic Math/Lagrange Multiplier.md):

$$\begin{aligned} &\text{Minimize} \qquad f(\omega, b, \alpha) = \frac{\|\omega\|^2}{2} + \sum_{n=1}^{N} \alpha_n (1 - y_n(\omega^T \mathbf{x}_n + b)) \\ &\text{subject to} \qquad \alpha_n \geq 0, \quad \forall\, n = 1, \ldots, N \end{aligned}$$

By taking the partial derivative, we get:

$$\frac{\partial(\frac{1}{2}\|\omega\|^2)}{\partial \omega} = \omega = \sum_{n=1}^{N} \alpha_n y_n \mathbf{x}_n,$$
$$\frac{\partial(\frac{1}{2}\|\omega\|^2)}{\partial b} = 0 = -\sum_{n=1}^{N} \alpha_n y_n$$

Substituting our above result into $f(\omega, b, \alpha)$, we get:

$$\begin{aligned} f_D(\omega, b, \alpha) &= \frac{1}{2}\omega^T\omega + \sum_{n=1}^{N}\alpha_n - \sum_{n=1}^{N}\alpha_n y_n \omega^T \mathbf{x}_n - \sum_{n=1}^{N}\alpha_n y_n b \\ &= \frac{1}{2}(\sum_{n=1}^{N}\alpha_n y_n \mathbf{x}_n)^T(\sum_{m=1}^{N}\alpha_m y_m \mathbf{x}_m) + \sum_{n=1}^{N}\alpha_n - \sum_{n=1}^{N}\alpha_n y_n (\sum_{m=1}^{N}\alpha_m y_m \mathbf{x}_m)^T \mathbf{x}_n \\ &= \frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N}\alpha_n \alpha_m y_n y_m \mathbf{x}_n^T \mathbf{x}_m + \sum_{n=1}^{N}\alpha_n - \sum_{n=1}^{N}\sum_{m=1}^{N}\alpha_n \alpha_m y_n y_m \mathbf{x}_m^T \mathbf{x}_n \end{aligned}$$

$= \text{sum}_{n=1}^{N}\alpha_n - \frac{1}{2}\sum_{n=1}^{N}\sum_{m=1}^{N}\alpha_n\alpha_m y_n y_m \mathbf{x}_n^T\mathbf{x}_m Now, we know we just need to maximize f_D$ under condition $\sum_{n=1}^{N}\alpha_n y_n = 0$ and $\alpha_n \geq 0$, which can be done with 'Lagrange Multiplier' again.

After we get the solution, we have:

$$\mathbf{w} = \sum_{n=1}^{N}\alpha_n y_n \mathbf{x}_n$$

and:

$$b = -\frac{1}{2}\left(\min_{n:y_n=+1}\mathbf{w}^T\mathbf{x}_n + \max_{n:y_n=-1}\mathbf{w}^T\mathbf{x}_n\right)$$

So, we can using above equations to solve ω and b.

Result We can check the result, and superisingly we find most of α_i are zero and α_i is non-zero if and only if $\mathbf{x}_n$ lies on one of the two bounaries (plus-plane and minus-plane). And since they are so speical and can impact our decision boundary a lot, we call them **"support vector"**. It means vector supports the margin boundaries.